ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** straight line	Straight.
2	**(d)** image	Reflection.
3	**(a)** reflection	Equal.
4	**(b)** luminous object	Gives light.
5	**(a)** reflection	Reflection.
6	**(d)** normal	Perpendicular.
7	**(d)** non-luminous	Reflects light.
8	**(c)** erect	Upright.
9	**(d)** incident ray	Falling.
10	**(c)** transparent	Clear.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Gives its own light (1); E.g. the Sun (1).	1+1
12	A line perpendicular to the surface (1); At the point of incidence (1).	1+1
13	Glass (1); Water (1).	1+1
14	Does not give light (1); E.g. the Moon (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Sun (1); Candle (1); Bulb (1).	1+1+1
16	Erect (1); Same size (1); Virtual (1).	1+1+1
17	No light (1); Objects do not reflect (1); Need light to see (1).	1+1+1
	OR: Moon (1); Table (1); Book (1).	1+1+1
18	Angle of incidence = angle of reflection (1); Rays and normal in one plane (1); Both (1).	1+1+1
	OR: Blocks more rays (1); Larger shadow (1); Closer to the source (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Luminous give light (1); Non-luminous do not (1); Sun vs Moon (1); Seen by own light vs reflected (1); Examples (1).	1+1+1+1+1
	OR: Three cards (1); Holes aligned (1); Candle on one side (1); See the light (1); Shift one card: no light (1).	1+1+1+1+1
20	Take a lamp (1); Hold an opaque object (1); Place a screen (1); Shadow forms (1); Change distance (1).	1+1+1+1+1
	OR: Opaque object (1); Light source (1); Screen (1); Size changes (1); Dark (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
- Writing that a pinhole camera forms an erect image or that a large hole gives a sharper picture.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** object	Same size.
2	**(a)** reflected ray	Returns.
3	**(c)** opaque	Blocks light.
4	**(c)** right	Lateral inversion.
5	**(a)** 20°	Equal.
6	**(c)** translucent	Partly.
7	**(c)** two mirrors	Mirrors.
8	**(a)** 60°	Equal.
9	**(d)** opaque objects	Block light.
10	**(a)** small hole	Pinhole.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A box with a tiny hole (1); Forms an image (1).	1+1
12	Wood (1); Brick (1).	1+1
13	Dark area behind an opaque object (1); Formed by blocking light (1).	1+1
14	An instrument with two mirrors (1); To see over things (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Glass (1); Water (1); Air (1).	1+1+1
16	Box (1); Small hole (1); Screen (1).	1+1+1
17	Submarines (1); Trenches (1); Over crowds (1).	1+1+1
	OR: Wood (1); Metal (1); Stone (1).	1+1+1
18	It reflects sunlight (1); Light reaches our eyes (1); It is non-luminous (1).	1+1+1
	OR: Reads correct in mirror (1); Drivers notice (1); Give way (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Virtual (1); Erect (1); Same size (1); Same distance (1); Laterally inverted (1).	1+1+1+1+1
	OR: Regular: smooth (1); Diffused: rough (1); Mirror (1); Paper (1); Both follow the law (1).	1+1+1+1+1
20	Take a box (1); Make a hole on one side (1); Put a screen at the other end (1); Point at an object (1); See inverted image (1).	1+1+1+1+1
	OR: Light from source (1); Reflect from object (1); Enter eye (1); Brain forms image (1); Need light (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
- Writing that a pinhole camera forms an erect image or that a large hole gives a sharper picture.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Light: Shadows and Reflections** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1 m	Equal.
2	**(b)** dark	Absence of light.
3	**(d)** upside down	Inverted.
4	**(a)** 3 m	Equal.
5	**(a)** screen	Surface.
6	**(c)** patterns	Mirrors.
7	**(d)** diffused reflection	Scatter.
8	**(a)** in light	Need light.
9	**(d)** mirror	Polished.
10	**(a)** regular reflection	Mirror.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	An image (1); Virtual and erect (1).	1+1
12	Bouncing back of light (1); From a surface (1).	1+1
13	A tube with mirrors (1); Makes patterns (1).	1+1
14	Left-right reversal (1); In a mirror (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Dressing (1); Vehicles (1); Periscope (1).	1+1+1
16	Equal (1); 35° (1); Law of reflection (1).	1+1+1
17	A surface on which a shadow or image falls (1); Wall (1); Cloth (1).	1+1+1
	OR: Dark (1); Same shape (1); Needs screen (1).	1+1+1
18	Equal (1); 4 m (1); Behind the mirror (1).	1+1+1
	OR: Rays seem to come from behind (1); Not real (1); Cannot be on screen (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tube (1); Two mirrors (1); At 45° (1); Reflect twice (1); See over objects (1).	1+1+1+1+1
	OR: Mirrors (1); Beads (1); Reflections (1); Patterns (1); Rotate (1).	1+1+1+1+1
20	Mirror (1); Periscope (1); Kaleidoscope (1); Vehicle mirrors (1); Spectacles (1).	1+1+1+1+1
	OR: Mirror (1); Water surface (1); Polished metal (1); Glass window (1); Moon shine (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that the Moon is luminous; it only **reflects** sunlight.
- Writing that the image in a plane mirror is real; it is **virtual** and cannot be caught on a screen.
- Forgetting that the distance between a person and the image is **twice** the distance from the mirror.
- Saying that a transparent object forms a dark shadow; only **opaque** objects form dark shadows.
- Measuring the angles from the mirror surface instead of from the **normal**.
- Writing that a pinhole camera forms an erect image or that a large hole gives a sharper picture.

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